Balark Tiwari

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Summary

Ph.D. researcher in Mechanical Engineering with 4+ years developing C, C++, and Python software for robotics, real-time control, and intelligent manufacturing. Experience includes robot kinematics/dynamics, trajectory generation, digital and state-space control, and system identification on Linux and NI Real-Time platforms. Built and integrated end-to-end robotic workcell software, from modeling and control implementation to safety monitoring and validation. Co-authored peer-reviewed publications and a filed patent on digital forming systems.

Education

University of Notre Dame — **Notre Dame, United States**
Ph.D. in Aerospace and Mechanical Engineering, Robotics and Dynamics — 2021-Present

University of Notre Dame — **Notre Dame, United States**
M.S. in Aerospace and Mechanical Engineering, GPA: 3.8/4 — 2021-2024

Delhi Technological University — **New Delhi, India**
B.Tech. in Engineering Physics, GPA: 8.6/10 — 2014-2018

Technical Skills

Programming and Platforms: C, C++ (multi-threaded, Linux), Python, CUDA, PyTorch, MATLAB/Simulink, ROS/ROS2, OpenCV, NI Real-Time, Linux (Ubuntu, `rt-preempt`)
Robotics & Controls: Forward and inverse kinematics, Jacobians, trajectory planning, robot dynamics and computed-torque control, motion planning, classical and modern servo control systems, digital control systems, Iterative Learning Control (ILC), feedback/feedforward control, closed-loop control, state estimation, sensor fusion, system identification, human-in-the-loop control
AI & Perception: Vision-Language Models (VLMs), Reinforcement Learning (Q-learning), LSTMs, CNNs, computer vision (segmentation, tracking, 3D reconstruction), Bayesian filtering
Hardware, Test, and Design Tools: Industrial robots and motion systems (multi-axis forming cell, EtherCAT drives), thermal and vision sensors, IMUs and force sensors, SolidWorks, LabVIEW, NI DAQ, automated calibration and verification routines, Gazebo, Prusa Slicer, Bambu Studio

Experience

Graduate Research Assistant - Robotics — 2021 - Present
University of Notre Dame - Advanced Manufacturing Laboratory — *Notre Dame, United States*

- Led full-stack software/hardware integration for a multi-axis digital glass/metal forming workcell; delivered a deterministic C/C++/`rt-preempt` Linux stack for repeatable operation and debugging (GitHub).
- Implemented reinforcement learning (Q-learning) for digital glass forming to achieve desired filament angle and workzone size/location, improving stability in edge-case scenarios and recovery from failure modes.
- Built multi-threaded C/C++ and Python architecture for synchronized multi-sensor DAQ (thermal/vision cameras, encoders, confocal sensor), with kHz-rate motion-control loops and sub-ms timing jitter on real-time threads.
- Implemented z-domain state-space tracking and ILC feedforward, reducing temperature-tracking RMSE by 40%.
- Used offline Gemini-based VLM inference to log print context and flag anomalies, supporting operator-in-the-loop recovery in failure-prone regimes.
- Collaborated with cross-functional lab peers (mechanical design, optics/metrology, process and controls) to integrate and validate subsystem improvements; co-authored a patent and peer-reviewed publications.

Graduate Teaching Assistant - Mechanical Engineering — 2021 - 2024
University of Notre Dame, Department of Aerospace and Mechanical Engineering — *Notre Dame, United States*

- Mentored students in hands-on projects on servo control, motion interpolation, and trajectory planning, delivering practical experience in sensor integration and real-time control system implementation.
- Supported instructors in preparing assignments, exams, lectures, and course materials for undergraduate courses in Mechanics, Analytical Dynamics, Differential Equations, Vibrations, and Controls, while also grading and hosting office hours.

Research Analyst, Strategy & Marketing — 2018 - 2019
The Smart Cube (WNS Global Services) — *Delhi, India*

- Conducted industry profiling, competitive landscape analysis, and market trend studies for US-based consulting firms and Fortune 100 clients, delivering concise, data-driven recommendations for senior stakeholders.

Selected Projects

Digital Glass Forming | *Real-Time Linux, Trajectory Control, ILC, VLM*

- Built and deployed a multi-threaded C++ `rt-preempt` stack for synchronized multi-sensor acquisition and kHz-rate motion control on a multi-axis robotic forming system.
- Replaced a baseline PID with a z-domain tracking controller on identified reduced-order thermal dynamics, using Internal Model Principle for asymptotic rejection of constant disturbances; augmented with feedback/feedforward and spatial-horizon ILC (non-causal learning) feedforward to compensate repeatable layer-to-layer disturbances, reducing temperature-tracking RMSE by 40%.

User Intent Recognition for EksoGT Exoskeletons | *Human-in-the-Loop Control, Sensor Fusion, PyTorch, LSTMs*

- Evaluated performance against a linear baseline and used feature-importance analysis to identify key temporal patterns and explain performance gaps for able-bodies and non-able-bodied subjects.

Digital Metal Forming | *Computer Vision, ILC, EtherCAT, Real-Time Control*

- Built an OpenCV pipeline for tool-deflection and part-shape estimation from camera and laser-line data to support in-process decision making.
- Formulated pass-to-pass ILC updates using force/motion feedback to reduce springback and improve geometric consistency.
- Integrated EtherCAT drives and mixed digital/analog I/O sensing for deterministic closed-loop machine control.

KUKA iiwa 7-DOF Manipulator Control | *Forward/Inverse Kinematics, Trajectory Planning, Computed-Torque Control*

- Modeled a 7-DOF KUKA iiwa arm using URDF and Denavit-Hartenberg parameters; implemented forward and inverse kinematics and Jacobian-based numerical inverse kinematics in SE(3) for workspace trajectory tracking.
- Designed smooth joint-space trajectories from task-space waypoints and implemented a computed-torque (model-based) controller using full robot dynamics (mass matrix, Coriolis, gravity) from MATLAB's Robotics System Toolbox to accurately track complex 3D paths.
- Validated performance in simulation with 3D visualizations of end-effector motion and joint behavior, iterating on controller gains and trajectory profiles to improve tracking accuracy and robustness.

Selected Publications

- **Research Impact Award 2026**, Institute for Control, Optimization and Networks (ICON) Conference, Purdue University.
- **PCT International Patent Application** - "Systems and Methods of Printing 3D Lattices" (Application No. PCT/US2026/010502, filed January 7, 2026; University of Notre Dame assignee). Patent pending.
- **Development of a Digital Metal Forming System**, ASME International Symposium on Flexible Automation.
- **Real Time Vision-Based Morphology Tracking of Glass Additive Manufacturing Processes**, SFF Symposium.